

Andrew Thompson

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EDUCATION

- **Ph.D. in Mechanical Engineering**, Northwestern University, Evanston, IL *Expected Dec 2025*
 - Advisor: Prof. Brenna Argall
- **Master of Science in Robotics**, Northwestern University, Evanston, IL *Dec 2019*
- **BS in Physics**, Indiana University, Bloomington, IN *May 2017*
- **BA in Folklore & Ethnomusicology**, Indiana University, Bloomington, IN *May 2017*

TECHNICAL SKILLS

- **Languages:** C/C++/C#, Python, Mathematica, Matlab, JavaScript, React, LaTeX
- **Software:** ROS/ROS2, OpenGL, OpenCV, PyTorch, Matlab, rviz, Autodesk, Unity, Linux Systems
- **Hardware:** 3D Printing, Prototyping, Precision Measurements and Drafting, Machining
- **Specializations:** Reinforcement Learning, Generative Models, Optimization and Planning, Signal Processing, Dynamics, Computer Vision, Human-Robot Interaction

RELEVANT EXPERIENCE

PhD Candidate

Sept 2020 – Ongoing

Northwestern University and Shirley Ryan AbilityLab (argallab), Chicago, IL

Body-Machine Interface (BoMI) Project

- Deployed generative BoMI system enabling real-time 6-DOF robotic arm teleoperation for individuals with cervical-level SCI (N=10), using wearable IMUs and ROS2.
- Learned user-customized motion encoders and control-space mappings using statistical modeling and novel optimization techniques in Python/C++; these control mappings do not require mode-switching.
- Analyzed skill acquisition and control adaptation using custom signal processing pipelines including GUI feedback (OpenGL).
- Led IRB-approved longitudinal human-subject study (12 sessions per participant); designed protocols, collected data, and coordinated with clinicians. Conducted 120 sessions from full study, 190+ sessions including scoping and vetting sessions.
- Wrote ROS2 wrapper (C++) for x-io x-IMU3 sensors; WIP to make this an officially supported package.

Bias-Aware Interface Remapping

- Designed algorithm to remap sparse joystick input into full-rank control signals using convex hull expansion and statistical analysis of user behavior (Python).
- Deployed with 3-Axis joystick in IRB-approved study (N=8: SCI, stroke); improved interface access without mode switching. Assessment includes a custom GUI for collecting interface usage data (OpenGL).

Information-Theoretic Nudging

- Developed real time, mutual information-based shared autonomy controller to resolve intent ambiguity during human-robot collaboration tasks using ROS and Python.
- Deployed controller on Kinova Jaco Gen2 arm; system adaptively nudged users toward high-certainty control states using automatic mode switching to reduce cognitive workload during teleoperation.

Meta-Participation and End-User Interviews

- Conducted structured interviews with motor-impaired robotics researchers/creators (N=3); extracted qualitative barriers to participation in research and proposed design guidelines.

Design Cycle Iterations for a Zero-Force Assistive Interface (WIP)

- Designed a series of virtual tasks for assessing interface usage (Unity with ROS2 TCP back-end).
- Conducted multiple rounds of iterative design cycles with members of the larger motor deficit community (ALS, SCI, Stroke) and industry partners (Function, LUCI) for the development of a novel zero-force optical sensor.

Post-Baccalaureate Researcher

Jan 2020 – Sept 2020

Northwestern University and Shirley Ryan AbilityLab (argallab), Chicago, IL

- Collected virtual wheelchair driving behavior data (N=12) for data-driven control modeling (OpenGL, React, JavaScript).
- Built and trained LSTM-based predictor to classify driving maneuvers from user input, kinematic trajectories (Python/PyTorch).
- Analyzed and compared predicted motion primitives against motor-impaired datasets from prior studies to infer intended commands for powered wheelchair driving.

Graduate Researcher, MSR Program Projects*Sept 2018 – Dec 2019**Northwestern University, Evanston, IL*

- Designed and fabricated a novel, cable-driven series-elastic actuator for back-drivable actuation.
- Built Bluetooth-controlled hexapod robot and designed a bio-inspired gait controller using embedded firmware.
- Built prototype ECG acquisition device and validated signal fidelity via analog signal processing and filtering.
- Wrote simulation for a bipedal rappelling robot with environmental constraints in Mathematica.
- Co-developed a hybrid fluidic actuator combining McKibben muscle and MMVS elements; implemented pneumatic manifold.
- Programmed Baxter robot to play interactive Pong using ROS and ultrasonic range sensors.

Summer Research Intern, Robotics and Embedded Systems (REMS)*June 2019 – Aug 2019**BlueHalo (formerly Intelligent Automation, Inc.), Rockville, MD*

- Contributed to haptics and dynamics research projects in REMS group; focused on multimodal sensor fusion, user-aware teleoperation, and real-time signal processing for teleoperation of large vehicles.
- Lead sub-thrust for R&D to avoid vehicle rollovers; daily agile development meetings and weekly technical presentations.

Lab Resource Manager and Research Assistant, de Ruyter Biophysics Lab*May 2016 – Aug 2018**Indiana University, Bloomington, IN*

- Maintained C. Vicina colonies and supported neurophysiology research targeting insect visual systems.
- Used C++ and Matlab to generate 1000+ Hz experimental stimuli and automate statistical analysis workflows.
- Updated and digitized internal training materials for onboarding and procedural consistency.

PUBLICATION(S) AND PATENT(S)

- USPTO Patent: SYSTEM AND METHOD FOR PROVIDING TACTILE FEEDBACK FOR PREVENTING ROLLOVER OF A VEHICLE, 2024, 18/804804
- NeurIPS '25 (IN REVIEW): Structured Semi-Supervised Generative Methods for Learning Robust Control Embeddings from Human Motion Data. **(1st Author)**
- ICORR '25 LBA: Quantifying User Adaptation Through Embedded Control Metrics in BoMI Systems. **(1st Author)**
- ICORR '25: A Customizable Fitting Protocol for the Body-Machine Interface. **(2nd Author)**
- HRI '24 Companion: An Evolution of Assistive Robot Control to Meet End-User Ability. **(1st Author)**
- HRI '24 A3DE Workshop: Navigating Adaptive Design: Advancing the Body-Machine Interface for 6d Control. **(2nd Author)**
- HRI '24 A3DE Workshop: Principal Component Analysis in Accessible Control. **(2nd Author)**
- RSS '23 Workshop: Identifying Accessibility Barriers to Robotics Research. **(1st Author)**
- Algorithmic Foundations of Robotics '22: Information Theoretic Intent Disambiguation via Contextual Nudges for Assistive Shared Control. **(2nd Author)**
- ICORR '22: Control Interface Remapping for Bias-Aware Assistive Teleoperation. **(1st Author)**

SERVICE AND OUTREACH

RSS 2025: Human-Robot Contact and Manipulation (HRCM)*October 2024-June 2025**Workshop Organizer*

- Co-authored the workshop proposal, co-designed the website, and reached out to various members of the research community for speaking roles. Also reviewed manuscript submissions.

ICORR 2025*May 2025**Volunteer*

- Served on a panel discussion for a symposium on collaboration between clinical staff and researchers.
- Performed various volunteering duties at the 2025 International Conference on Rehabilitation Robotics in Chicago.
- Conducted guided tours of partner organization Shirley Ryan AbilityLab, hosted visitors with robotics demonstrations.

Northwestern University*September 2021 – December 2022**Teaching Assistant and Grader*

- ME 499: Robotic Manipulation (Fall 2022), ECE 301: Introductory Robotics Lab (Winter 2022), ECE 495: Machine Learning & Artificial Intelligence for Robotics (Fall 2021)

Museum of Science and Industry*April 2019 – April 2025**National Robotics Week: Robot Block Party Demos*

- Facilitated a variety of interactive, educational robotics demonstrations for Chicago-area community members. These demos involved wearable robotic devices, teleoperation of robotic arms and roboticized wheelchairs, and assistive control blending.